

Application of Image Processing

Module 2: Medical Image Processing: Concepts and Applications

Radiology

It is a branch of medicine that uses medical imaging techniques to diagnose and sometimes treat diseases inside the human body. It is essentially the “medical eye” that allows doctors to see inside without surgery.

Radiology relies on imaging modalities that generate images of internal organs, tissues, and structures. These images are then analyzed either by radiologists (doctors specializing in imaging) or by medical image processing systems.

Major imaging modalities in radiology:

1. X-ray (Radiography)
 - Produces 2D shadow images.
 - Good for bones, fractures, lung infections.
2. Computed Tomography (CT Scan)
 - Multiple X-rays from different angles → 3D cross-sectional images.
 - Used for brain, chest, abdomen imaging.
3. Magnetic Resonance Imaging (MRI)
 - Uses strong magnetic fields and radio waves.
 - Excellent for soft tissues (brain, muscles, joints).
4. Ultrasound (Sonography)
 - Uses sound waves.
 - Common in pregnancy imaging, abdominal organs, heart (echocardiography).
5. Nuclear Medicine (PET / SPECT)
 - Uses radioactive tracers.
 - Shows organ function and metabolism.

Medical image processing is applied to radiological images to improve diagnosis and research. Some key roles of image processing in radiology:

- Image Enhancement: Improve contrast, remove noise (e.g., making tumors more visible).
- Segmentation: Automatically separating organs, tissues, or abnormalities (e.g., lung nodule detection).
- Registration: Aligning images from different times/modalities (e.g., pre- and post-surgery CT scans).
- 3D Reconstruction: Creating 3D models from CT/MRI scans.
- Computer-Aided Diagnosis (CAD): AI/ML models that help radiologists detect diseases faster (e.g., breast cancer in mammograms, pneumonia in chest X-rays).
- Quantitative Analysis: Measuring tumor size, bone density, blood flow, etc.

Why is Radiology Important in Medical Image Processing?

- Radiology provides the raw data (images).
- Medical image processing provides tools to interpret, analyze, and assist diagnosis.
- Together, they make diagnoses faster, more accurate, and less invasive.

Diagnostic Testing

Diagnostic testing in medicine refers to procedures and techniques used to identify, confirm, or rule out diseases and medical conditions. It helps doctors move from “symptoms” → “confirmed diagnosis” so that they can plan treatment.

Types of Diagnostic Testing: Diagnostic tests can be broadly divided into:

1. Medical Imaging Tests (Radiology-based)

- Purpose: To see inside the body non-invasively.
- Examples:
 - X-ray → bones, chest, fractures.
 - CT Scan → tumors, brain injuries, internal bleeding.
 - MRI → brain, spinal cord, soft tissues.
 - Ultrasound → pregnancy monitoring, abdominal organs.
 - PET Scan → cancer spread, metabolic activity.

2. Laboratory Tests

- Purpose: To analyze body fluids or tissues.
- Examples:
 - Blood tests (CBC, glucose, cholesterol).
 - Urine tests.
 - Biopsy (microscopic examination of tissue).

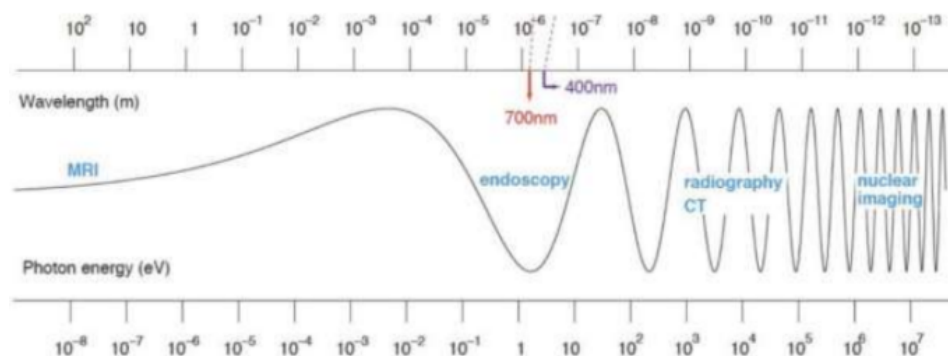
3. Physiological / Functional Tests

- Purpose: To measure how well an organ is working.
- Examples:
 - ECG (Electrocardiogram) → heart's electrical activity.
 - EEG (Electroencephalogram) → brain activity.
 - Pulmonary function test → lungs' capacity.

Role of Medical Imaging in Diagnostic Testing: Medical imaging is a subset of diagnostic testing. It provides a visual diagnosis without cutting open the body.

- Early detection → e.g., tumors, fractures, internal bleeding.
- Treatment planning → e.g., deciding where to operate.
- Monitoring progress → e.g., seeing if cancer shrinks after chemotherapy.

Electromagnetic Spectrum



The electromagnetic (EM) spectrum represents the range of all types of electromagnetic radiation, organized by wavelength or photon energy. Different parts of the spectrum are used in medical imaging technologies depending on their ability to penetrate tissues and provide diagnostic information.

Key Regions of the Spectrum in Medical Imaging

1. Radio Waves (102 m to ~10⁻¹ m) – Used in MRI
 - Magnetic Resonance Imaging (MRI) relies on radio waves and strong magnetic fields.
 - It works by detecting the response of hydrogen nuclei in the body when placed in a magnetic field.
 - Advantages:
 - Excellent soft tissue contrast
 - Non-invasive

- No ionizing radiation
- 2. Microwaves / Millimeter Waves (~10–3 m)
 - Rarely used in medical imaging, but explored in experimental applications like hyperthermia treatment and some forms of diagnostic imaging.
- 3. Infrared to Visible Light (10–6 m to 10–7 m) – Used in Endoscopy
 - Endoscopy uses visible and near-infrared light (wavelengths around 400–700 nm).
 - Optical fibers transmit light into body cavities to visualize tissues directly.
 - Advantages:
 - Real-time imaging
 - Useful for procedures involving the gastrointestinal tract, respiratory system, etc.
- 4. X-rays (~10–11 m to 10–13 m) – Used in Radiography and CT
 - Radiography (X-ray imaging): Uses X-rays to image bone structures and lungs.
 - Caution: Exposure to ionizing radiation, requiring controlled usage.
 - CT scans (Computed Tomography): Uses X-rays in multiple directions to create cross-sectional images.
 - Advantages:
 - High spatial resolution
 - Fast acquisition times
- 5. Gamma Rays (~10–13 m and smaller) – Used in Nuclear Imaging
 - Nuclear imaging (PET, SPECT): Uses gamma rays emitted by radioactive tracers injected into the body.
 - Provides metabolic and functional information, such as identifying tumors or blood flow abnormalities.
 - Advantages:
 - Detects physiological processes
 - High sensitivity
 - Caution: Involves radioactive substances requiring careful handling.

Positron Emission Tomography: PET is a type of nuclear medicine imaging that uses radioactive substances (radioisotopes) which emit positrons to create detailed images of processes inside the body. Unlike CT or MRI (which mainly show structure), PET shows function and metabolism — how tissues and organs are actually working.

How PET Works (Step by Step)

1. Injection of a Radiotracer
 - A small amount of radioactive material is injected into the bloodstream.
 - The tracer travels through the body and collects in highly active cells (like cancer cells)
2. Positron Emission
 - The tracer decays and emits positrons (positive electrons).
 - When a positron meets an electron in the body, they annihilate each other → releasing two gamma rays moving in opposite directions.
3. Detection by PET Scanner
 - The PET scanner detects these gamma rays.
 - By analyzing where they came from, the computer reconstructs an image showing areas of tracer concentration.
4. Image Formation
 - Areas with more tracer uptake = brighter spots → these usually indicate high metabolic activity

Medical Uses of PET

- Cancer Detection & Monitoring: Detects tumors, their spread (metastasis), and response to treatment.
- Heart Disease: Shows blood flow and damaged heart muscle after a heart attack.
- Brain Disorders: Alzheimer's disease, epilepsy, Parkinson's, brain tumors.

Nuclear Medicine: Nuclear medicine is a medical specialty that utilizes radioactive substances (radionuclide or radio-tracer), known as radiopharmaceuticals, for diagnosis and treatment of various diseases. It focuses on assessing bodily functions and detecting disease at a molecular level, often in its early stages.

Diagnosis: Nuclear medicine imaging, like SPECT and PET scans, allows doctors to track the movement of radioactive tracers within the body to visualize organ function and identify abnormalities.

Treatment: Radiopharmaceuticals can be used to target and destroy cancerous cells or to treat specific conditions like hyperthyroidism.

Radioactive Tracers:

- The tracers are designed to be absorbed by specific organs or tissues, allowing for targeted imaging and treatment.
- Uses radioisotopes (Radioisotopes are the unstable form of an element that emit radiation to transform into a more stable form) to provide images of how tissues and organs function.
- The radioisotope produces radiation and is attached to a chemical that is absorbed by the body. As the radioisotope emits radiation, a special camera and computer detect it and produce an image.
- Used to diagnose cancer, investigate blood circulation and evaluate organ disorders.
- Also used for treating some cancers like prostate, thyroid or breast cancer.

Radiation Protection

Medical imaging techniques like X-rays, CT scans, and fluoroscopy use ionizing radiation to create images. While these are very useful for diagnosis and treatment, ionizing radiation can damage body cells and increase cancer risk if not controlled. Radiation protection is therefore the set of strategies, techniques, and principles used to minimize unnecessary radiation exposure while still obtaining high-quality diagnostic images.

Dosimetry (Radiation Dose): Dosimetry is the science of measuring how much radiation is absorbed by the body.

Types of Dosimetry:

- Physical Dosimetry
 - Ionization chambers → measure radiation by collecting charges produced in air.
 - Photodosimetry → uses light emitted by certain materials when exposed to radiation.
 - Thermoluminescent dosimeters (TLDs) → small crystals that trap energy and release light when heated, used to measure cumulative dose.
- Patient Dosimetry
 - Measures the actual dose received by the patient during a procedure.
 - Important for monitoring exposure in repeated scans (e.g., cancer follow-ups).

Radiation Safety Principles: These are the fundamental rules to keep radiation dose as low as possible:

Time: Minimize exposure time → the less time a patient or staff is exposed, the lower the dose.

Distance:

- Inverse Square Law: Radiation intensity decreases rapidly with distance.
- Example: Standing 2 meters away instead of 1 meter reduces exposure by a factor of 4.

Shielding

- Use of protective barriers between the source and the body.
- Common shielding tools:

- Lead aprons → protect torso.
- Thyroid shields → protect thyroid gland.
- Lead barriers/walls → protect staff in X-ray rooms.

Radiation Optimization: It means using the lowest possible dose while still achieving diagnostic-quality images.

Justification of Radiation Use

- Every imaging test involving radiation should be clinically justified.
- Doctors must ensure that:
 - Benefits outweigh risks.
 - No alternative (e.g., MRI or ultrasound, which don't use ionizing radiation) is available.

Molecular Imaging: Molecular imaging is an advanced form of medical imaging that focuses on visualizing, characterizing, and measuring biological processes at the molecular and cellular levels inside the body, often before structural changes (like tumors) are visible. Unlike conventional imaging (X-ray, CT, MRI), which mainly shows anatomy/structure, molecular imaging shows:

- Function (how cells/tissues work)
- Biochemistry (enzymes, receptors, metabolism)

Cellular changes that happen in early disease stages

How Molecular Imaging Works

1. Molecular Probes (Tracers/Agents)
 - Special molecules (often radioactive or fluorescent) are injected into the body.
 - These probes are designed to bind to specific cellular targets (like cancer cell receptors).
2. Imaging Technology
 - The probes emit signals (radioactive, optical, or magnetic).
 - Imaging devices (like PET, SPECT, Optical Imaging, MRI) detect these signals.
3. Image Formation
 - A computer processes the signals to create images showing molecular activity in tissues and organs.

Example: PET in Cancer Detection

 - PET (Positron Emission Tomography) is the most common molecular imaging technique.
 - Uses a tracer called FDG (fluorodeoxyglucose), which behaves like glucose.
 - Cancer cells consume more glucose → tracer accumulates in tumors.
 - PET images highlight these areas as bright spots, allowing detection of cancer at an early stage.

Applications of Molecular Imaging

- Oncology (Cancer): Detects tumors, metastasis, and treatment response.
- Neurology: Detect Alzheimer's disease (amyloid plaques), Parkinson's, epilepsy.
- Cardiology: Assess blood flow, detect damaged heart tissue after heart attacks.
- Drug Development: Test how new drugs interact with specific targets in the body.

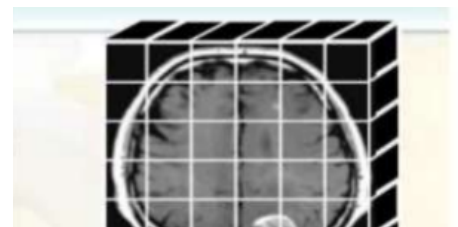
Medical Image Representation - Voxels

When medical images move from 2D to 3D, the concept of a voxel (volume element) comes in. A voxel is the smallest 3D unit of an image, just like a pixel is the smallest 2D unit.

Voxel as a 3D Element: A voxel represents a tiny rectangular box of tissue in the patient's body. Its dimensions are determined by:

- Pixel size (width × height in-plane, mm²).
- Slice thickness (depth along the third axis, mm).

Example: If pixel size = 1 × 1 mm² and slice thickness = 5 mm, then each voxel = 1 × 1 × 5 mm³.



How Voxel Dimensions Are Defined: The voxel volume is determined by:

Field of View (FOV):

- The area of the body being imaged, measured in mm (or cm).
- Larger FOV = larger pixel size = larger voxels.

Pixel Size:

- Derived from $\text{FOV} \div \text{number of pixels in the image matrix}$.
- Smaller pixels → smaller voxels → higher resolution.

Slice Thickness:

- The depth of each slice (the "z-axis" dimension).
- Thinner slices → smaller voxels → finer 3D detail.

Voxel Intensity

- Each voxel is assigned a numerical value representing the average signal intensity of all tissue within that volume.
- In MRI, this depends on proton density, relaxation times (T1/T2), etc.
- In CT, this corresponds to Hounsfield Units (HU).

Why Voxels Are Important

- 3D Imaging: Voxel representation allows 3D reconstructions, enabling doctors to view anatomy from different angles.
- Volume Measurements: Voxel size determines the accuracy of measuring organ or tumor volumes.
- Resolution Trade-offs:
 - Small voxels → higher resolution, but require longer scan time and more data storage.
 - Large voxels → faster scans, but less detail (partial volume effects).

$$\text{pixel size} = \frac{\text{FOV}}{\text{matrix}} \times \frac{\text{FOV}}{\text{matrix}} \quad (\text{mm}^2) \quad \text{voxel size} = \frac{\text{FOV}}{\text{matrix}} \times \frac{\text{FOV}}{\text{matrix}} \times \text{thickness} \quad (\text{mm}^3)$$

Grayscale representation

- A grayscale image stores one value per pixel representing luminance, commonly 8 bits per pixel (256 shades), though higher bit depths (10–16 bits) provide more tonal detail for scientific and medical imaging.
- Perception-wise, grayscale corresponds to the black–white axis; in RGB terms, equal channel values
- $R=G=B$
- $R=G=B$ lie on the cube diagonal from black to white, forming the gray axis.

Color representation (RGB and others)

- RGB encodes color with three channels Red, Green, Blue; typical 8-bit per channel “24-bit color” yields ~16.7 million colors, with the gray axis where
- $R=G=B$
- $R=G=B$ producing neutrals.
- Other models include CMY/CMYK for printing (subtractive mixing) and perceptual spaces like HSV/HSL; devices and workflows choose models suited to emission vs print and manipulation tasks.

Bit depth and “depth–color”

- Bit depth per channel determines the number of discrete intensity steps: an n-bit channel supports 2^n levels; total combinations depend on channels (e.g., 8 bpc RGB ≈ 16.7M colors).
- Higher bit depth reduces banding and preserves gradients/edges; grayscale bit depth similarly controls the number of gray shades (e.g., 8-bit = 256, 12-bit = 4096).

Lookup tables (LUTs/CLUTs)

- A color lookup table (CLUT) maps stored pixel indices or intensities to display RGB values; indexed color images store indices into a 256×3 table instead of full RGB per pixel.
- In scientific imaging, LUTs map numeric intensities to grayscale or pseudo-color for visualization; contrast limits define linear mapping between min–max to brightness with clipping outside range.

Grayscale vs pseudo-color via LUTs

- A grayscale LUT maps intensity to a single-channel luminance rendering; preferred for faithful intensity interpretation and accessibility, with pseudo-color used to highlight ranges or label IDs when dynamic range is high.
- Pseudo-color assigns different hues to intensity bands, aiding feature detection but requiring clear legend and awareness that the LUT changes appearance, not the underlying data.

Practical implications

- Choose representation by task: store and process at higher bit depth to avoid banding; display with appropriate LUT and visible min–max to prevent clipping.
- For displays and apps, RGB with 8 bpc plus optional alpha is standard; indexed color with CLUT is used to reduce memory/bandwidth or for specific visualization pipelines.

Medical Imaging File Formats

DICOM (Digital Imaging and Communications in Medicine)

- Purpose: The most widely used standard in medical imaging (CT, MRI, PET, ultrasound, etc.).
- Structure:
 - Contains both image data and metadata (patient info, study description, acquisition parameters)
 - Stores multiple images (slices) as part of one study.
- Advantages:
 - Universally accepted in hospitals.
 - Ensures interoperability between scanners,
 - Secure (supports encryption and anonymization).
- Disadvantages:
 - Large file size due to embedded metadata.
 - Complex standard; harder to work with directly compared to research-oriented formats.

Analyze 7.5

- Purpose: An older format, widely used for storing 3D biomedical images (especially MRI and CT) for research.
- Structure: Two separate files:
 1. .hdr (Header) → stores metadata (image dimensions, voxel size, orientation).
 2. .img (Image) → stores the raw voxel intensity values.
- Advantages:
 - Simple and compact format.
 - Was a standard in neuroimaging before NIfTI.
- Disadvantages:
 - Does not handle complex metadata as well as DICOM.
 - Separate files (.hdr and .img) can get mismatched or lost.

NIfTI (Neuroimaging Informatics Technology Initiative)

- Purpose: A modern format developed for neuroimaging research (fMRI, DTI, MRI).
- Structure: Can store in two ways:
 - Single file (.nii) → header + image data combined.
 - Dual file (.hdr + .img) → similar to Analyze 7.5, but extended.
- Advantages:
 - Stores spatial orientation and transformations → crucial for brain mapping.
 - Supports time-series data (important for fMRI).
 - Widely used in research software (SPM, FSL, AFNI).
- Disadvantages:
 - Not widely supported in hospitals compared to DICOM.
 - Mostly research-focused.

Interfile

- Purpose: Mainly used in nuclear medicine (PET, SPECT).
- Structure:
 - ASCII-based header file (.hdr) → human-readable text describing acquisition parameters.
 - Binary image data file (.img) → contains pixel/voxel values.
- Advantages:
 - Flexible and easy to read/edit because the header is in text form.
 - Suitable for custom research and reconstruction algorithms.
- Disadvantages:
 - Not standardized as strictly as DICOM.
 - Less common outside nuclear medicine.

Signal-to-Noise Ratio (SNR)

SNR is influenced by sensor quality, exposure settings, quantum efficiency, and environmental factors. In low-light imaging, photon shot noise becomes a major limiting factor for SNR. Improving SNR can involve increasing exposure time, using better optics, enhancing sensor sensitivity, or applying noise reduction algorithms. High SNR is essential in scientific imaging, surveillance, and any domain where fine details are critical. SNR measures the strength of the useful signal relative to background noise:

$$SNR = \frac{\text{Standard Deviation of Noise}}{\text{Mean Signal}}$$

- Impact: Low SNR images appear grainy and obscure fine details, limiting diagnostic or analytical accuracy.
- Improvement Strategies: Longer exposure times, cooling sensors, averaging multiple frames, and advanced denoising algorithms.
- Applications: MRI, astrophotography, remote sensing, scientific imaging.

Image Segmentation: Image Segmentation is finding the group of pixels that are similar.

- It is partitioning the image into various subgroups (of pixels) called Image Objects.
- Assigning labels to pixels, and the pixels with the same label fall under a single category.
- It can reduce the complexity of the image so analysing the image becomes simpler.
- Using these labels, users can specify boundaries, draw lines, and separate the most required objects in an image from unnecessary pixels.

Algorithms detect instances, which provide information about individual objects, and hence all the persons have different colors. Why Image Segmentation is Needed?

1. Identify the objects from the image.
2. Not only identifying, Also Provides more insight information about the objects.

Applications

- Facial recognition - identifying an employee to mark attendance automatically.
- Medical industry - faster diagnosis, detecting diseases, tumors, cell and tissue patterns from radiography, MRI, endoscopy, thermography, ultrasonography, etc.

Types of Image Segmentation Techniques based on the image properties:

1. Similarity Detection (Region based Approach)
 - Detecting similar pixels in an image based on a threshold, region growing, region spreading, and region merging.
 - Machine learning algorithms like clustering relies on this approach of similarity detection on an unknown set of features. It detects similarity based on a pre-defined (known) set of features.
2. Discontinuity Detection (Boundary / Edge based Approach)
 - Searches for discontinuity.
 - Here, edges are detected based on various metrics of discontinuity like intensity, color etc.
 - E.g. Image Segmentation Algorithms like Edge Detection, Point Detection, Line Detection

Threshold based Segmentation

Threshold value T is considered as a constant. But that approach may be ineffective considering the amount of noise of the image. So, we can either keep it constant or change it dynamically based on the image properties to obtain better results.

- Global Threshold - If we want to divide the image into two regions (object and background), we define a single threshold value.
- Local Threshold - If we have multiple objects along with the background, we must define multiple thresholds. These thresholds are collectively known as local threshold.

Based on that, thresholding is of the following types:

1. Simple Thresholding
 - This technique replaces the pixels in an image with either black or white.
 - If the intensity of a pixel (I_{ij}) at position (i, j) is less than the threshold (T), then replace that with black and if it is more, then we replace that pixel with white. This is a binary approach to thresholding.
2. Otsu's Binarization
 - In global thresholding, we had used an arbitrary value for threshold value and it remains a constant.
 - How can we define and determine the correctness of the selected threshold? A simpler but rather inept method is to trial and see the error.
 - Take an image histogram that has two peaks (bimodal image), one for the background and one for the foreground.
 - According to Otsu binarization, for that image, we can approximately take a value in the middle of those peaks as the threshold value.
 - So, it automatically calculates a threshold value from an image histogram for a bimodal image.
 - The disadvantage is for images that are not bimodal, the image histogram has multiple peaks, or one of the classes (peaks) present has high variance.
3. Adaptive Thresholding
 - A global value as threshold value may not be good in all the conditions where an image has different background and foreground lighting conditions in different actionable areas.
 - We need an adaptive approach that can change the threshold for various components of the image.
 - In this, the algorithm divides the image into various smaller portions and calculates the threshold for those portions of the image.
 - Hence, we obtain different thresholds for different regions of the same image.
 - This in turn gives us better results for images with varying illumination.

- The algorithm can automatically calculate the threshold value.
- The threshold value can be the mean of neighborhood area or it can be the weighted sum of neighborhood values where weights are a Gaussian window (a window function to define regions).

Edge based Segmentation

- Edge detection is the process of locating edges in an image based on various discontinuities in grey level, colour, texture, brightness, saturation, contrast etc.
- To further enhance the results, supplementary processing steps must be followed to concatenate all the edges into edge chains that correspond better with borders in the image.
- Edges consist of meaningful features and contain significant information.
- It significantly reduces the size of the image and filters out information that may be regarded as less relevant, preserving and focusing solely on the important structural properties of an image for a business problem.

Edge detection algorithms are two categories:

- Gradient based methods and Gray Histograms.
- Basic edge detection operators like sobel operator, canny, Robert's variable etc.
- These operators aid in detecting the edge discontinuities and hence mark the edge boundaries.
- The end goal is to reach at least a partial segmentation using this process, where we group all the local edges into a new binary image where only edge chains that match the required existing objects or image parts are present.

Region based Segmentation

Creating segments by dividing the image into various components having similar characteristics. Region-based image segmentation techniques initially search for some seed points – either smaller parts or considerably bigger chunks in the input image. Next, either add more pixels to the seed points or further diminish or shrink the seed point to smaller segments, and merge with other smaller seed points.

Region Growing

- It's a bottom to up method where you begin with a smaller set of pixel and start accumulating or iteratively merging it based on certain pre-determined similarity constraints.
- If there is a match or similarity in neighboring pixels, then they are added to the initial seed pixel, thus increasing the size of the region.
- When we reach saturation, the growth of that region cannot proceed further. So, the algorithm now chooses another seed pixel, which necessarily does not belong to any region(s) that currently exists and start the process again.
- Region growing methods often achieve effective Segmentation that corresponds well to the observed edges.
- But sometimes, when the algorithm lets a region grow completely before trying other seeds, that usually biases the segmentation in favour of the regions which are segmented first.
- To counter this effect, most of the algorithms begin with the user inputs of similarities first, no single region is allowed to dominate and grow completely and multiple regions are allowed to grow simultaneously.
- Region growth, also a pixel based algorithm like thresholding but the major difference is thresholding extracts a large region based out of similar pixels, from anywhere in the image whereas region-growth extracts only the adjacent pixels.
- Region growing techniques are preferable for noisy images, where it is highly difficult to detect the edges.

Region Splitting and Merging

- The splitting and merging based segmentation methods use two basic techniques done together in conjunction – region splitting and region merging – for segmenting an image.
- A region split, unlike the region growth, considers the entire input image as the area of business interest.

- Then, it would try matching a known set of parameters or pre-defined similarity constraints and picks up all the pixel areas matching the criteria.
- This is a divide and conquers method as opposed to the region growth algorithm.

WATERSHED BASED SEGMENTATION

- It considers the analogy of geographic landscape with ridges and valleys for various components of an image. The slope and elevation of the topography are distinctly quantified by the gray values of the respective pixels – called the gradient magnitude.
- Based on this 3D representation which is usually followed for Earth landscapes, the watershed transform decomposes an image into regions that are called “catchment basins”.

For each local minimum, a catchment basin comprises all pixels whose path of steepest descent of gray values terminates at this minimum. The algorithm considers the pixels as a “local topography” (elevation), often initializing itself from user-defined markers. Then, the algorithm defines “basins” which are the minima points and hence, basins are flooded from the markers until basins meet on watershed lines.

The watersheds are formed that separate basins from each other. Hence the picture gets decomposed because we have pixels assigned to each such region or watershed

Clustering based Segmentation: Clustering is dividing the population (data points) into a number of groups, such that data points in the same groups are more similar to other data points in that same group than those in other groups (clusters).

k-means clustering: The k represents the number of clusters.

1. First, randomly select k initial clusters
2. Randomly assign each data point to any one of the k clusters
3. Calculate the centers of these clusters
4. Calculate the distance of all the points from the center of each cluster
5. Depending on this distance, the points are reassigned to the nearest cluster
6. Calculate the center of the newly formed clusters
7. Finally, repeat steps (4), (5) and (6) until either the center of the clusters does not change or we reach the set number of iterations

The key advantage of using the k-means algorithm is that it is simple and easy to understand.

FEATURE EXTRACTION: Feature extraction is the process of transforming raw data into a simplified and informative set of features or attributes. This reduces data complexity and highlights the most relevant information making it easier for machine learning models to analyze and learn from the data efficiently. It plays an important role in improving model accuracy and reducing computational costs by focusing on essential aspects of the data.

Feature extraction is **important** for several reasons:

- **Reduced Computation Cost:** Raw data, especially from images or large datasets can be very complex. Feature extraction makes this data simpler hence reducing the computational resources needed for processing.
- **Improved Model Performance:** By focusing on key features, machine learning models can work with more relevant information leading to better performance and more accurate results.
- **Better Insights:** Reducing the number of features helps algorithms concentrate on the most important data, eliminating noise and irrelevant information which can lead to deeper insights.
- **Prevention of Overfitting:** Models with too many features may become too specific to the training data, making them perform poorly on new data. Feature extraction reduces this risk by simplifying the model.

There are various techniques for extracting meaningful features from different types of data:

1. **Statistical Methods:** Statistical methods are used in feature extraction to summarize and explain patterns of data.

Common data attributes include:

- Mean: The average value of a dataset.
- Median: The middle value when it is sorted in ascending order.
- Standard Deviation: A measure of the spread or dispersion of a sample.
- Correlation and Covariance: Measures of the linear relationship between two or more factors.
- Regression Analysis: way to model the link b/w a dependent variable and one or more independent factors.

These statistical methods can be used to represent the center trend, spread and links within a collection.

2. **Dimensionality Reduction:** Dimensionality reduction reduces the number of features without losing important information. Some popular methods are:

- Principal Component Analysis: It selects variables that account for most of the data's variation, simplifying the dataset by focusing on the most important components.
- Linear Discriminant Analysis (LDA): It finds the best combination of features to separate different classes, maximizing class separability for better classification.
- t-Distributed Stochastic Neighbor Embedding (t-SNE): A technique that reduces high-dimensional data into two or three dimensions ideal for visualizing complex datasets.

3. **Feature Extraction for Textual Data:** In Natural Language Processing (NLP), we often convert raw text into a format that machine learning models can understand. Some common techniques are:

- Bag of Words (BoW): Represents a document by counting word frequencies, ignoring word order, useful for basic text classification.
- Term Frequency-Inverse Document Frequency (TF-IDF): Adjusts word importance based on frequency in a specific document compared to all documents, highlighting unique terms.

4. **Signal Processing Methods:** It is used for analyzing time-series, audio and sensor data: Signal processing methods

- Fourier Transform: It converts a signal from the time domain to the frequency domain to analyze its frequency components.
- Wavelet Transform: It analyzes signals that vary over time, offering both time and frequency information for non-stationary signals.

5. **Image Data Extraction**

- Histogram of Oriented Gradients (HOG): This technique finds the distribution of intensity gradients or edge directions in an image. It's used in object detection and recognition tasks.
- Convolutional Neural Networks (CNN) Features: They learn hierarchical features from images through layers of convolutions, ideal for classification and detection tasks.

Image Classification

Statistical Approach:

Based on probability and statistical models of pixel values or extracted features. The idea is: each class (e.g., object type) has a probability distribution, and new images are classified by finding the class with the highest likelihood.

Examples:

- Bayesian Classifier (Naïve Bayes) → assumes independence between features, assigns class with max posterior probability.
- K-Nearest Neighbors (KNN) → compares image feature vectors with training samples using distance measures (Euclidean, Mahalanobis).
- Support Vector Machines (SVM) → finds a hyperplane separating classes based on statistical optimization.

Advantages:

- Well-understood theory, simple to implement.
- Works well with structured data and limited samples.

Disadvantages:

- Relies heavily on feature extraction.
- Performance drops with high-dimensional or complex data.

Rule-Based Approach

- Uses explicit rules defined by experts or heuristics to classify images.
- Relies on feature engineering (e.g., shape, color, texture) and if-then rules.

Example: In medical imaging: Rule: "If the circularity > 0.9 and the average intensity $< \text{threshold}$ $\rightarrow$ classify as tumor."

Techniques Used:

- Thresholding (global or adaptive)
- Region-growing methods
- Decision trees with manually set conditions

Advantages:

- Simple, interpretable (human-understandable).
- Useful when domain knowledge is strong.

Disadvantages:

- Not robust for complex or noisy images.
- Hard to scale—rules may not generalize well.

Neural Network Approach

- Inspired by the human brain—uses layers of interconnected "neurons."
- Learn directly from raw data or extracted features, automatically finding patterns.

Types:

- Shallow Neural Networks $\rightarrow$ classical MLP (Multi-layer Perceptron), used in early image recognition.
- Convolutional Neural Networks (CNNs) $\rightarrow$ modern standard for image classification:
 - Convolution layers extract features (edges, shapes, textures).
 - Pooling layers reduce dimensionality.
 - Fully connected layers perform final classification.

Advantages:

- High accuracy with large datasets.
- Automatically learns relevant features (less manual feature engineering).
- Scales well to complex real-world problems (e.g., medical imaging, self-driving).

Disadvantages:

- Requires large training data and high computational power.
- Can be a "black box" (less interpretable).
- Risk of overfitting if not properly regularized.

Image Registration

Image alignment, or image registration, is the process of overlaying images of the same scene taken at different times, from different viewpoints, or using different sensors to achieve spatial correspondence. The goal is to align these images such that they correspond to the same spatial coordinates. This is crucial in various fields such as medical imaging, remote sensing, and computer vision, where precise alignment of images is required for accurate analysis and interpretation.

Temporal Alignment: Aligning images taken at different times to track changes over time.

Spatial Alignment: Aligning images captured from different viewpoints or sensors to create a unified view.

Multimodal Alignment: Aligning images taken with different imaging modalities (e.g., MRI and CT scans)

Imagine you take two photos of the same person:

- One from the front
- One from the side

To compare or combine them, you need to align them properly so that the head, eyes, or body match up. That process is called image registration.

In medical imaging, for example, doctors may want to align a CT scan (shows bones) with an MRI scan (shows soft tissues) of the same patient to get a complete picture.

1. Rigid Body Registration

- What it means: Only moves the whole image by shifting (translation) and rotating.
- Think of it like moving or rotating a toy car without bending or changing its shape.
- No stretching, no shrinking.
- Use case: Aligning two brain scans where the brain shape doesn't change.

2. Principal Axis Registration

- What it means: Uses the main directions of the object inside the image to align them.
- Example: If you scan a banana from two angles, the banana's long axis is the same in both images. Aligning those axes helps match the images.
- Works well when the object has a clear main orientation (like bones, elongated organs, or shapes).

3. Interactive Principal Axis Registration

- What it means: Same as above, but here the user helps the computer by adjusting manually.
- Example: A radiologist may rotate or shift an image by hand if the computer's automatic alignment isn't perfect.
- Good for situations where human judgment is better than automation.

4. Feature-Based Registration

- What it means: Instead of using the whole image, it looks for key points/features (like edges, corners, landmarks).
- Example: Matching the corners of a building in two satellite photos or matching the eyes/nose in two face photos.
- In medical imaging: matching bones, blood vessels, or organ boundaries.
- Advantage: Works well even if images are from different machines (CT vs MRI).

5. Elastic (Non-Rigid) Deformation Registration

- What it means: Unlike rigid registration, here the image can bend, stretch, or compress to match another image.
- Example: If you take a CT scan of lungs while breathing in and another while breathing out, the shape changes.
- Elastic registration helps match them by deforming one image to fit the other.
- Useful in tracking tumors or organ movement.

Rigid-Body Transformation – Mathematics: Rigid-body transformation means moving an object without changing its shape or size. That means only rotation and translation (no scaling, no shearing, no bending).

1. Components of Rigid-Body Transformation

1. Translation

- o Shifting the image in x,y(and maybe z) direction.
- o Example: Moving an object 5 units to the right and 2 units up.
- o Formula (2D): where t_x, t_y are translation amounts.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

2. Rotation

- o Rotating the image around the origin (or a chosen point).
- o Formula (2D rotation by angle θ):

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

3. Combination (Rotation + Translation)

- o Usually, both are applied together:
- o This is the rigid-body transformation equation in 2D.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

2. Homogeneous Coordinates (Matrix Form): To make things easier, we use homogeneous coordinates (add a 1 at the end of vectors). Then both translation and rotation can be written in a single matrix multiplication.

- The 3×3 matrix is called the transformation matrix.
- It combines both rotation and translation in one step.
- Works the same way in 3D, but the matrix becomes 4×4 .

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

3. Properties of Rigid-Body Transformation

1. Preserves distances

- o The distance between any two points before and after transformation remains the same. $\|p_1 - p_2\| = \|p'_1 - p'_2\|$

2. Preserves angles

- o The angle between two lines or vectors does not change.

3. Orthogonal Rotation Matrix

- o The rotation part RRR has a special property: $R^T R = I, \det(R) = 1$
- o This ensures no stretching happens.

4. Example (2D Case): Suppose a point $p=(2,1)$. We want to rotate it by 90° counterclockwise and then translate by $(3,2)$.

1. Rotation matrix for 90° :

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

2. Apply rotation:

$$\begin{bmatrix} x_r \\ y_r \end{bmatrix} = R \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

3. Apply translation:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 + 3 \\ 2 + 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

Final transformed point = $(2,4)$.

5. Extension to 3D:

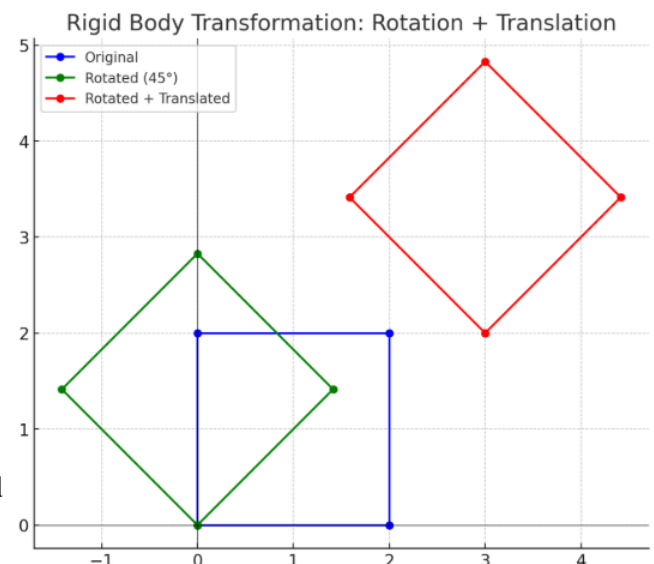
- Same concept, but now points are (x,y,z) .
- The transformation matrix is 4×4 .
- Includes rotation around x, y, z axes and translation (t_x, t_y, t_z) .

● Blue = Original square

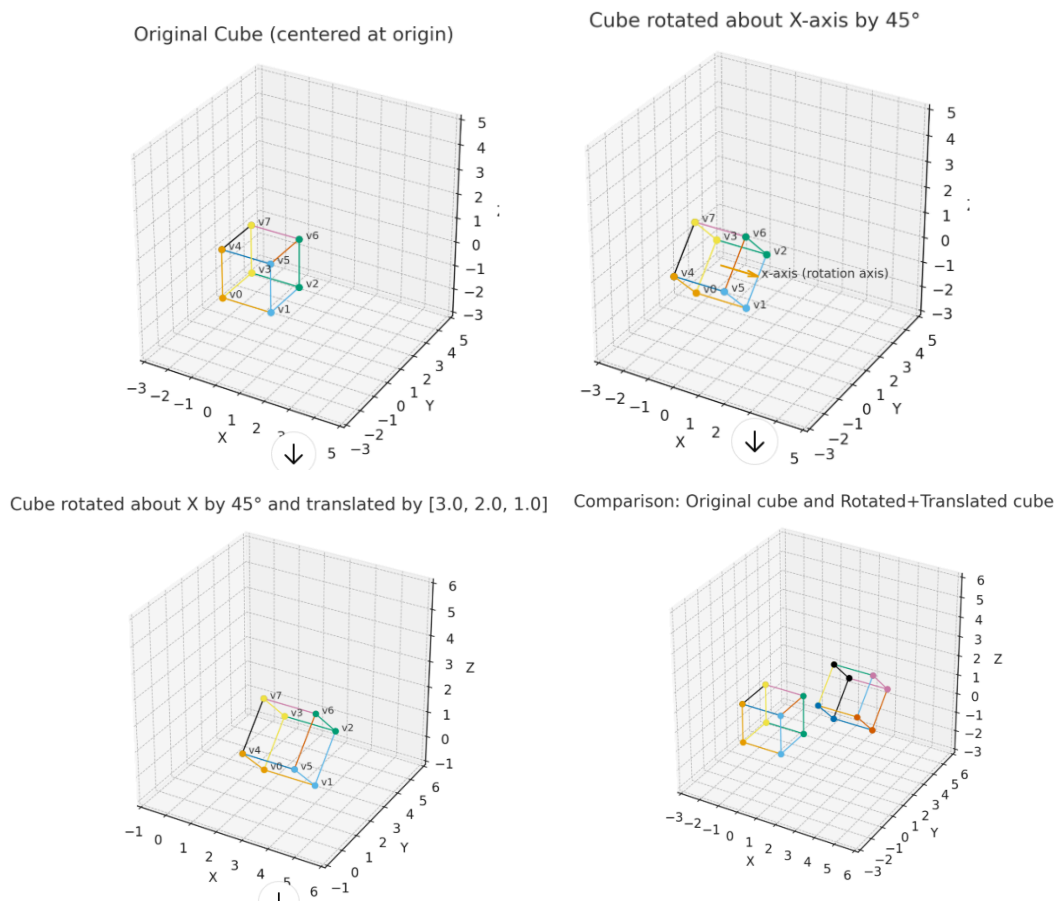
● Green = After 45° rotation (around the origin)

● Red = After rotation + translation (shifted right & up)

Notice how the shape and size are preserved — only the position and orientation change.



A step-by-step 3D visualization showed what happens when a cube is rotated about the x-axis and then translated. You should see four 3D plots (displayed above) and a numeric example.



1. Figure 1 — Original cube
 - o Cube centered at the origin (vertices labeled v0..v7). Side length = 2 (so coordinates ± 1).
 - o This is the starting shape.
2. Figure 2 — Cube rotated about the x-axis by 45°
 - o Rotation axis is the x-axis (an arrow marks it).
 - o Notice each vertex's x coordinate stays the same, while its (y,z) coordinates rotate in the yz-plane.
3. Figure 3 — Rotated then translated
 - o After rotating, the cube was translated by the vector $t = [3, 2, 1]$.
 - o This shows how rotation + translation moves the cube in space without changing its shape.
4. Figure 4 — Overlay comparison
 - o Original cube and the rotated+translated cube plotted together so you can see the net displacement.

Feature-Based Methods

SIFT (Scale-Invariant Feature Transform): SIFT is a robust method for detecting and describing local features in images. It identifies key points that are invariant to scale and rotation, making it effective in matching features across images with varying sizes and orientations.

Key Steps:

- **Keypoint Detection:** Identifies distinctive points in the image.
- **Feature Description:** Generates descriptors for the keypoints that capture the local image region's appearance.

Applications: Object recognition, image stitching, and panorama creation.

SURF (Speeded-Up Robust Features): SURF is an accelerated alternative to SIFT, designed to be faster while maintaining similar robustness. It uses integral images to speed up the computation of the detector and descriptor.

Key Steps:

- Keypoint Detection: Detects keypoints using a Hessian matrix-based approach.
- Feature Description: Uses a descriptor based on the distribution of pixel intensities around the keypoints.

Applications: Real-time object recognition, video tracking.

ORB (Oriented FAST and Rotated BRIEF): ORB combines the FAST keypoint detector with the BRIEF descriptor, providing a fast and efficient feature extraction method. It includes orientation information to improve robustness to rotation.

Key Steps:

- Keypoint Detection: Uses the FAST algorithm with orientation information.
- Feature Description: Uses the BRIEF descriptor, which is binary and computationally efficient.

Applications: Real-time applications, such as robotics and augmented reality.

RANSAC (Random Sample Consensus): RANSAC is a robust algorithm used for estimating transformation models while handling outliers. It is often used in conjunction with feature-based methods to estimate the best transformation between images.

Key Steps:

- Random Sampling: Selects random subsets of data to estimate the transformation.
- Model Fitting: Computes the transformation model and evaluates its quality.
- Outlier Detection: Identifies and excludes outliers to improve the robustness of the model.

Applications: Image stitching, 3D reconstruction.

Elastic deformation-based registration

Elastic deformation-based registration is a type of deformable image registration (DIR) that uses principles of mechanical elasticity to align two or more images. This non-rigid method is used when images have complex local variations that cannot be aligned with simpler transformations like rotation or translation.

The process of elastic registration is formulated as an optimization problem that minimizes a cost function. The core elements are:

1. Image Similarity Measure: A metric to quantify the difference between the fixed (reference) image and the moving image, after it has been deformed. Common metrics include:
 - o Sum of Squared Differences (SSD)
 - o Mean Squared Distance (MSD)
 - o (Normalized) Mutual Information (MI/NMI), which is useful for registering images from different modalities (e.g., MRI and CT).
2. Regularization: A constraint is added to the optimization process to ensure the deformation field is smooth and anatomically plausible. In elastic registration, this term models the deformation as an elastic material, such as a thin plate or elastic body.
3. Optimization: An algorithm is used to find the optimal deformation that minimizes the cost function. Common methods include gradient descent and multi-resolution approaches, which start with a coarse registration and are progressively refined.

Common algorithms and techniques

- **Finite Element Methods (FEM):** This approach treats the images as elastic bodies and applies non-uniform meshes to important features. It effectively models large local deformations but is computationally intensive.
- **Spline-Based Registration:** This is a popular method that uses piecewise continuous functions, such as B-splines or thin-plate splines, to model the deformation. B-splines offer local control over the deformation, meaning changes in one area do not affect distant parts of the image.
- **Demons Algorithm:** In this algorithm, the moving image is seen as a fluid or elastic body, and forces ("demons") are generated by local intensity differences that push the floating image toward the target image. It is computationally fast and often used in conjunction with other methods.
- **Inelastic Deformable Image Registration (i-DIR):** This is an extension of the elastic model that can handle discontinuous motion, like organ sliding, without requiring prior knowledge of where the discontinuities occur. It does this by incorporating an inelastic regularization term.

Applications: Elastic registration is particularly useful in medical imaging for applications where tissues and organs deform and change shape.

- **Adaptive Radiation Therapy (ART):** Used to track organ motion and anatomical changes during a course of treatment, allowing for adjustments to radiation plans. This is crucial for organs that shift with breathing, such as the lungs.
- **Image-Guided Surgery:** Enables surgeons to track localized deformations of soft tissue (e.g., in the brain, heart, or liver) in real-time during a procedure.
- **Longitudinal Studies:** Allows researchers to quantify changes in anatomy over time by registering images taken at different points, useful for studying conditions like Alzheimer's or evaluating disease progression.
- **Multimodality Fusion:** Combines images from different scanners, such as fusing functional PET images with anatomical CT or MRI images to provide a more complete clinical picture.
- **Biomechanical Analysis:** Used to estimate and analyze the strain and deformation of organs like the lungs by registering images taken at different points in the respiratory cycle.

Advantages:

- **Handles complex, local deformations:** Unlike rigid or affine methods, elastic registration can accurately model non-linear shape variations.
- **Ensures plausibility:** The regularization term helps prevent non-physical transformations, ensuring the deformation is smooth and anatomically reasonable.
- **Versatile:** Can be applied in many clinical scenarios, from tracking organ motion to analyzing disease progression.

Challenges:

- **Computational cost:** Many elastic algorithms, especially finite-element and B-spline models, can be computationally intensive and slow for high-resolution images.
- **Validation difficulty:** It can be challenging to quantitatively validate the accuracy of deformable registration, as "ground truth" deformations in living tissue are difficult to measure.
- **Parameter tuning:** Optimal results often depend on carefully tuning parameters like the weighting parameter (λ), which can vary by application and image type.
- **Discontinuities:** Standard elastic models can produce distorted results near discontinuities, like sliding organ boundaries, though advanced methods like i-DIR have been developed to address this.

Image Visualization: Once images are registered (aligned), doctors and researchers need to see them in useful ways.

1. 2D Display Methods

- Show images slice by slice or in different planes.

- Examples:
 - o Axial view: from the top of the head looking down.
 - o Coronal view: from the front.
 - o Sagittal view: from the side.
- Overlay visualization: Place two images on top of each other, e.g., showing PET scan (activity) on top of MRI (structure).

2. 3D Display Methods

- Instead of flat slices, build a 3D model of the body or organ.
- Surface rendering: Show only surfaces (like skin or bones).
- Volume rendering: Show the entire 3D volume with transparency (you can see inside).
- Use case: Surgeons can look at a 3D heart model before operating.

3. Virtual Reality (VR) Based Interactive Visualization

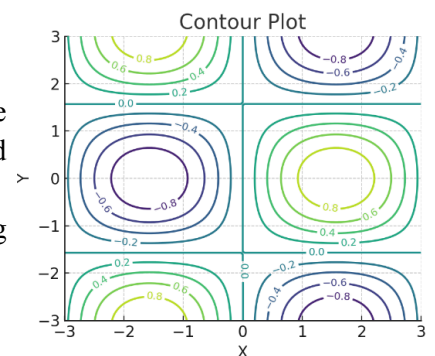
- Put on VR goggles → step into the body in 3D.
- Doctors can walk around, zoom in, rotate, and even “cut” through layers of tissue virtually.
- Example: Training medical students or planning a brain surgery with a realistic 3D brain model.

2D Image visualization: Display methods for visualization range from conventional 2D displays to immersive virtual reality (VR) environments, with each approach offering distinct benefits and use cases.

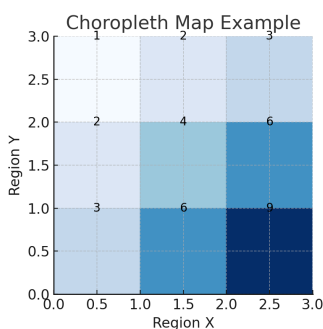
2D display methods: 2D visualization is the most traditional and widely used method, displaying data as static, two-dimensional images such as charts, maps, and photographs. While simpler and less expensive than 3D or VR methods, 2D displays can be less effective at conveying scale and proportion, often requiring multiple static images to show different perspectives.

Common 2D techniques

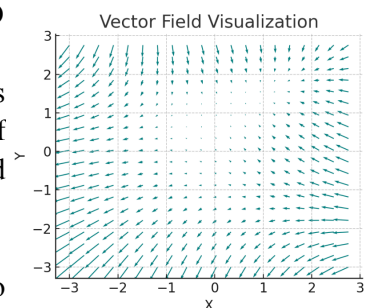
- Contour plots: These use contour lines to represent values of a third variable on a two-dimensional plane, commonly seen in geographic and meteorological maps.
- Scatter plots and line graphs: These are fundamental tools for displaying relationships between two variables.



- Heatmaps: These use color intensity to represent data density or intensity on a 2D surface.



- Vector field visualization: This technique uses icons or streamlines to show direction and magnitude of flow. Examples include line-integral convolution (LIC) and glyphs (icons) on a grid.

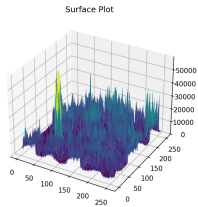


- Choropleth maps: These use color-coding to represent data variability within geographic areas.

3D display methods: 3D visualization uses computer software to create lifelike, three-dimensional images or animations that can be viewed on a 2D display, such as a monitor. This provides a better sense of depth and scale, but some areas of the model can be obscured by objects in the foreground.

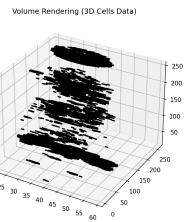
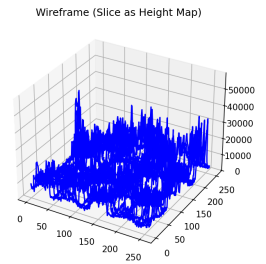
Common 3D techniques

- Wireframe models: These represent 3D objects as a skeletal framework of lines and vertices. This technique is computationally light and useful for understanding the underlying structure of a model.



- Surface plots: This method depicts a surface representing the relationship between two independent variables (X and Y) and a dependent variable (Z).

- Volume rendering: This technique visualizes a 3D dataset, such as a CT or MRI scan, by creating a semi-transparent view that allows internal structures to be seen.



- Photorealistic rendering: It is a process that uses advanced techniques like global illumination to create highly realistic static images that simulate the effects of light and shadow.
- 360° panoramas and virtual tours: These interactive visualizations allow users to explore a 3D scene from a fixed point by rotating and zooming the image. Multiple panoramas can be linked to create a full virtual tour.
- 3D animations (fly-throughs): These are sequences of rendered images played as a video to guide the viewer on a predetermined path through a space.

Virtual reality (VR) based interactive visualization: VR offers the most immersive and interactive experience, completely replacing the user's view of the real world with a computer-generated 3D environment. Users typically wear a head-mounted display (HMD) and use motion controllers to navigate and interact with the data spatially.

Key features and methods

- Full immersion: The user is fully enveloped in a virtual environment, with the visual experience adapting in real-time to their head and body movements.
- Spatial representation: VR transforms charts and graphs into 3D environments, allowing users to perceive trends and relationships spatially. For example, a complex data graph can become an explorable landscape.
- Interactive exploration: Users can intuitively manipulate and explore visualizations. Examples include zooming into data points, rotating complex models, or walking through a virtual architectural model.
- Collaborative analysis: Multiple users from different physical locations can meet in a shared virtual space to interact with data visualizations together in real-time, fostering teamwork and shared insights.
- Applications: VR is used across many fields, including scientific research, medical imaging for surgical planning, architectural design, and urban planning.
- VR hardware: Devices like HMDs, motion controllers, and haptic feedback gloves are used to enhance the sense of immersion and allow natural interaction with the virtual world.